Project Utilities & Features:

Algorithm Simulation: Executes non-preemptive Shortest Job First (SJF) and non-preemptive Priority Scheduling, utilizing custom tie-breaking logic (earlier arrival time wins).

Dynamic Gantt Chart Generation: Visually maps out the CPU timeline, rendering scaled blocks for each process execution phase and clearly marking CPU "Idle" periods.

Interactive Input & Strict Validation: Allows users to build custom workloads with process IDs, arrival times, burst times, and priority levels. The system includes robust validation to reject duplicate IDs, negative arrival times, zero/negative burst times, and non-numeric inputs.

Pre-loaded Test Scenarios: Features built-in workload scenarios to quickly demonstrate specific scheduling concepts:

- **Scenario A (Mixed Workload):** Standard test with varied arrivals and bursts.
- **Scenario B (Conflict):** Pits a short-burst/low-priority job against a long-burst/high-priority job.
- **Scenario C (Starvation):** Demonstrates how lower-priority jobs can be indefinitely delayed (starved) by continuous higher-priority arrivals.

Automated Comparison Engine: Displays results side-by-side and automatically calculates the "winner" for each metric (lowest average WT, TAT, and RT).

Analytical Conclusion Generation: Synthesizes the data to answer conceptual questions. It assesses if SJF actually favored short jobs, flags potential starvation risks in the Priority algorithm, evaluates the "fairer" algorithm based on maximum wait times, and provides a final recommendation on which algorithm suits the given workload best.